



Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

October 23, 2025

10 CFR 50.73

SVPLTR 25-0064

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Unit 3
Renewed Facility Operating License No. DPR-25
NRC Docket No. 50-249

Subject: Licensee Event Report 249/2024-002-02, "Low Pressure Coolant Injection Pump Suction Relief Valve Inoperable as a Primary Containment Isolation Valve Resulted in a Condition Prohibited by Technical Specifications"

Enclosed is Licensee Event Report 249/2024-002-02, "Low Pressure Coolant Injection Pump Suction Relief Valve Inoperable as a Primary Containment Isolation Valve Resulted in a Condition Prohibited by Technical Specifications." This report describes an event being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by Technical Specifications.

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Mr. Daniel J. Murphy, Regulatory Assurance Manager, at (779) 231-7443.

Respectfully,

A handwritten signature in black ink that reads "Hardik Patel".

Hardik Patel
Site Vice President
Dresden Nuclear Power Station

Enclosure: Licensee Event Report 249/2024-002-02



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10A), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Dresden Nuclear Power Station, Unit 3

☒ 050
☐ 052

2. Docket Number

00249

3. Page

1 OF 3

4. Title

Low Pressure Coolant Injection Pump Suction Relief Valve Inoperable as a Primary Containment Isolation Valve Resulted in a Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
11	20	2024	2024	- 002 -	02	10	23	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode

1 -- Power Operation

10. Power Level

100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Daniel J. Murphy, Site Regulatory Assurance Manager

Phone Number (Include area code)

(779) 231-7443

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	NH	RV	C568	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 11/20/2024 at 1407 CST, the Main Control Room was notified that the operating handle for the 3A Low Pressure Coolant Injection (LPCI) Suction Relief Valve was found positioned higher than the operating handle for the other relief valves in the system, making the valve inoperable as a Primary Containment Isolation Valve (PCIV). Technical Specification (TS) 3.6.1.3 Condition C was entered. At 1650 CST, the operating handle for the relief valve was moved to the down position in accordance with station procedure and TS 3.6.1.3 Condition C was exited. The valve had been replaced during refueling outage activities on 11/3/2024. At 0202 on 11/15/2024, Unit 3 entered Mode 2 and Limited Condition of Operation (LCO) 3.6.1.3 became applicable. LCO 3.6.1.3 Required Action C.1 directs isolation of the affected flow path within 4 hours. When the flow path was not isolated by 0602 on 11/15/2024, Required Actions E.1 and E.2 directed shutdown to Mode 3 within 12 hours and Mode 4 within 36 hours, respectively. Therefore, a condition prohibited by Ts existed by the failure to be in Mode 3 by 1802 on 11/15/2024, and Mode 4 by 1802 on 11/16/2024. This event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by TS.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Dresden Nuclear Power Station	☒ 050	2. DOCKET NUMBER 00249	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 02

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric – Boiling Water Reactor, 2957 megawatts thermal rated core power

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

A. CONDITIONS PRIOR TO EVENT

Unit: 3 Event Date: November 20, 2024 Event Time: 1407 CST

Reactor Mode: 1 Mode Name: Power Operation Power Level: 100%

No structures, systems or components were inoperable at the time of the event that contributed to the event.

B. DESCRIPTION OF EVENT

On 11/20/2024 at 1407 CST, with Dresden Unit 3 at 100 percent power, the Main Control Room was notified that the operating handle for the 3A Low Pressure Coolant Injection (LPCI) [BO] Suction Relief Valve [RV] was found position higher than the operating handles of the other relief valves in the system. The relief valve [RV] is normally seated closed and relieves back to the suppression pool [NH] to provide over-pressure protection of the LPCI [BO] piping when the pump [P] is isolated. The relief valve [RV] also functions as a Primary Containment Isolation Valve (PCIV) [ISV]. As a PCIV [ISV], its safety-related function is to isolate containment from the LPCI [BO] system. Technical Specification (TS) 3.6.1.3 Condition C requires lines with only one PCIV [ISV] to be isolated within 4 hours and to be verified isolated every 31 days when in Operating Modes 1, 2, and 3. If this action is not met, Condition E of TS 3.6.1.3 requires the operating unit to be placed in Mode 3 within 12 hours and Mode 4 within 36 hours. At 0202 on 11/15/2024, Unit 3 entered Mode 2 and Limited Condition of Operation (LCO) 3.6.1.3 became applicable. LCO 3.6.1.3 Required Action C.1 directs isolation of the affected flow path within 4 hours. When the flow path was not isolated by 0602 on 11/15/2024, Required Actions E.1 and E.2 directed shutdown to Mode 3 within 12 hours and Mode 4 within 36 hours, respectively. Therefore, a condition prohibited by TS existed by the failure to be in Mode 3 by 1802 on 11/15/2024, and Mode 4 by 1802 on 11/16/2024. The operating handle for the 3A LPCI [BO] Suction Relief Valve [RV] was moved to the down position in accordance with station procedure and TS 3.6.1.3 Condition C was exited at 1650 CST on 11/20/2024.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by Technical Specifications.

C. CAUSE OF EVENT

The relief valve [RV] had been replaced on 11/03/2024 during a refueling outage. Prior to the replacement of the relief valve [RV], there is no record of valve failure or mispositioning.

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CONTINUATION SHEET**

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	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 02

NARRATIVE

An evaluation of the event was completed. The cause was determined to be that the relief valve [RV] process does not provide instruction to verify relief valves [RV] are in the closed position prior to return to service.

D. SAFETY ANALYSIS

The safety significance of the event is very low. The open relief valve [RV] piping ties back to the same suction piping upstream of the 3A LPCI [BO] pump [P] suction. The open relief valve [RV] did not create any additional flow paths that would divert LPCI [BO] pump [P] discharge flow away from the reactor vessel or suppression pool [NH] and did not negatively affect the available pump [P] suction head. As discussed in the Dresden Updated Final Safety Analysis Report (UFSAR), for closed system piping which communicates with the suppression pool [NH] and is expected to remain submerged during a Loss of Coolant Accident (LOCA), the intact piping or the water seal acts as the penetration isolation barrier and ensures that the primary containment boundary is maintained intact until another barrier can be established to isolate the penetration. The LPCI [BO] suction line is closed system piping which communicates with the suppression pool [NH] and is expected to remain submerged during a LOCA. Therefore, the water seal can be credited as an isolation barrier maintaining the safety function of the primary containment system.

E. CORRECTIVE ACTIONS

At 1650 CST on 11/20/2024, the operating handle for the 3A LPCI [BO] Suction Relief Valve [RV] was moved to the correct (down) position in accordance with station procedure.

Additional corrective actions were developed by the evaluation of the event. These include actions to update relief valve [RV] planner guides and model work orders for IST Program Safety Related Relief Valves [RV] to include a sign off step to verify the valve is in the closed position after installation. These actions are tracked in the station's Corrective Action Program.

F. PREVIOUS OCCURENCES

A 5-year internal corrective action program and LER database review was performed for Dresden Nuclear Power Station (DNPS). One occurrence of a relief valve [RV] operating handle being found in the lifted position from DNPS from 2019 was identified. The cause of this event was determined to be that the work package instructions did not include direction for control of the relief valve [RV] lifting device during or upon completion of installation. The relief valve [RV] for this event was part of a different system and of a different type than the relief valve that is the subject of this report. However, the evaluation performed for the 2019 event had corrective actions which were closed without the necessary rigor to prevent recurrence; specifically, a thorough effectiveness review of the corrective actions from the 2019 event could have prevented the subject event of this report.

G. COMPONENT FAILURE DATA

No equipment failures are applicable.